Physical & Environmental Sciences
UNIVERSITY OF TORONTO
SCARBOROUGH

EESA11H: Environmental Pollution
Summer 2026
COURSE SYLLABUS
Starting May 4, 2026
OFFERED ONLINE (Quercus) ASYNCHRONOUS
Modules posted once per week on Mondays via Quercus

COURSE CONTACTS

Course Website: EESA11H3 Y LEC01 20265:Environmental Pollution

Course Instructor:
Dr. Kathy Wallace - kathy.wallace@utoronto.ca
Office hours: drop in via Zoom – times and meeting link on Quercus
Please email only with issues of personal/private nature (see note below)

Teaching Assistant:
Valentyna Kundas – v.kundas@mail.utoronto.ca
Office hours: drop in via Zoom – times and meeting link on Quercus

A note about emails: We kindly request that you NOT email the instructor or TA directly regarding questions about the course material, assignments, or exams. We have set up discussion boards for you to ask your questions along with your fellow students. Please reserve emails for personal matters only. Our TA and I would enjoy meeting you during our office hours to discuss the course and get to know you.

EESA11H Environmental Pollution (from UTSC Course Calendar)

"*This course illustrates the environmental effects of urban expansion, changing methods of agriculture, industrialization, recreation, resource extraction, energy needs and the devastation of war. Drawing on information from a wide spectrum of topics - such as waste disposal, tourism, the arctic, tropical forests and fisheries - it demonstrates what we know about how pollutants are produced, the pathways they take through the global environment and how we can measure them. The course will conclude with an examination of the state of health of Canada's environments highlighting areas where environmental contamination is the subject of public discussion and concern. No prior knowledge of environmental science is required*".

COURSE OBJECTIVES:

This course provides an overview of how environmental pollution has impacted our planet. We will examine the nature and causes of pollution and their impact on the environmental systems of the planet: Atmosphere, Lithosphere, Hydrosphere, and Biosphere. Topics will include air pollution, climate change, water-based and land-based pollution impacts. We will address the new emerging issues in environmental pollution including plastics, forever chemicals and nuclear waste. Canadian and international case studies will be presented to reinforce understanding of the issues affecting our environment.

By the end of this course students should be able to:

- Describe how humans and environmental pollutants are impacting our planet.
- Explain concepts and terms related to environmental pollution including earth systems, chemicals, and transport mechanisms.
- Understand the future implications of unrestrained pollution to our planet.

COURSE MATERIALS

There is no required text for this course, however we will have assigned readings of select scientific papers or articles in each week's Module. In addition, YouTube type videos will also be provided in most Modules. Each reading and video will have accompanying "read along or watch along" questions to guide you in your learning and understanding. The content of the readings and videos is course content, equally important as watching course lectures.

Recorded video lectures will be provided each week and posted in Quercus in the Module for that week. The lecture will include recorded videos by the instructor including a PowerPoint slide presentation like a traditional lecture. There will be occasional short YouTube type videos embedded within the lecture recording. These embedded videos are equally important as lecture content. A copy of the lecture slides used by the instructor will be provided as a PDF file for you to follow along and to facilitate your note taking.

In summary each Module will consist of:

- Lecture video recordings
- Lecture slides provided as PDF file
- Course reading(s) and/or Course Video
- Questions to assist with understanding of course readings and/or videos
- Summary of Module content, including key terms

All the above noted Module components are considered "examinable" course content for online quizzes, mid-term exams, and the final exam.

ASSIGNMENTS AND EVALUATION

Component	Evaluation	Date/Content	Value
1	Quiz 1	All content from Modules 1 to 3 Online via Quercus **Available May 26 @ 6pm to May 27 @ 10pm** **(begin no later than May 27 at 10pm)**	12.5%
2	Mid-Term Test	**All content from Modules 1 to 5** **Online via Quercus** **Thursday - June 11 available to begin** **between 5 to 10 pm (begin no later than 10pm)**	30%
3	Quiz 2	All content from Modules 6 to 8 **Online via Quercus** **July 14 @6pm to July 15 @ 10pm** **(begin no later than July 15 at 10pm)**	12.5%
4	Final Exam	The final exam is cumulative and will be **in person** at the UTSC campus during the final exam period, which is August 10 to August 22. Date to be scheduled by the registrar's office.	45%
		TOTAL	**100%**

Evaluation Outline

1. **and 3. On-line (Quercus) Quizzes 1 and 2 (12.5% each)**

QUIZ 1 – Open to start on Quercus May 26 @ 6pm to May 27 @ 10pm

QUIZ 2 – Open on Quercus July 14 @ 6pm to July 15 @ 10pm

Two online class quizzes (Quiz 1 and Quiz 2) will be given on the above noted dates. Quiz 1 will be focused on the content of Modules 1 to 3. Quiz 2 will focus on Modules 6 to 8. Students will have 60 minutes to complete the quiz. Questions will be a combination of multiple choice, true or false, and fill in the blank type questions. Although this test quiz is open book, sharing of answers or working with other students during the quiz is **not allowed and considered academic misconduct.**

3. **Mid-Term Exam (30%) – June 11 – Online Via Quercus – Available to start between 5 to 10pm – 120 mins – must begin no later than 10pm to have full 120 mins**

A two-hour (120 minute) ONLINE (Quercus) mid-term exam will be held online via Quercus on June 11. To accommodate your preparation for the mid-term test, there will be no Module presented during the week of June 8. Questions will be a combination of multiple choice, true or false, and fill in the blank type questions. **IMPORTANT**: If you miss the mid-term exam for a verifiable reason (such as documented illness), an attempt will be made to organize ONE make-up mid-term date at the discretion of the instructor. **You must submit any documentation* for the reason you missed the mid-term within 3 days after the missed mid-term (i.e. no later than 11:59 pm on June 15) to be "considered" for a make-up mid-term exam.** If you simply "miss" the mid-term exam, you will receive a mark of zero. The validity of missing the mid-term exam will be assessed directly by Dr. Wallace. This could result in a grade of zero if your absence was not acceptable. Unacceptable reasons for missing mid-term exam include travel, employment, lateness, and forgetfulness. *Accommodations outlined in a Letter of Accommodation, or a Letter of Consideration generated by UTSC AccessAbility Services will be respected.*

*If you miss the midterm exam, you must submit relevant paperwork BOTH through ACORN and through the DPES self-declaration absence form with supporting documentation to be considered to write a make-up midterm. Please refer to the following link: https://www.utsc.utoronto.ca/physsci/self-declaration-absence-form-0 . Note, for A-level courses such as this one, it is not permissible to transfer the value of a missed mid-term to the final exam. **You must submit your documentation within 3 days of the missed mid-term to be considered for a make-up midterm.**

4. **Final Exam (45%) – Date to be determined**

The two-hour (closed book) final exam will be cumulative, meaning that material from the entire course is subject to examination. All course material, which includes lecture recordings, lecture slides, readings, videos, as well as "read and watch along questions" is subject to examination. The final exam format will include multiple choice, true or false, and fill in the blank type questions. The final exam will be IN PERSON on the UTSC campus. The date of the exam will be during the final exam period (August 10 to August 22) and is scheduled by the registrar's office.

IMPORTANT INFORMATION

AccessAbility Services

Students with diverse learning styles and needs are welcome in this course. In particular, if you have a disability/health consideration that may require accommodations, please feel free to approach me and/or the **AccessAbility** Services Office as soon as possible. I will work with you and **AccessAbility** Services to ensure you can achieve your learning goals in this course. Inquiries are confidential. The University of Toronto is committed to accessibility. If you require accommodations for a disability, or have any accessibility concerns about the course, the classroom or course materials, please contact The UTSC Accessibility Services as soon as possible: http://www.utsc.utoronto.ca/~ability/

We also suggest you also refer to the following University of Toronto Scarborough Library link: http://utsc.library.utoronto.ca/services-persons-disabilities

Religious observances

The University provides reasonable accommodation of the needs of students who observe religious holy days other than those already accommodated by ordinary scheduling and statutory holidays. Students have a responsibility to alert members of the teaching staff in a timely fashion to upcoming religious observances and anticipated absences and instructors will make every reasonable effort to avoid scheduling tests, examinations or other compulsory activities at these times.

Please reach out to me as early as possible to communicate any anticipated absences related to religious observances, and to discuss any possible related implications for course work.

Equity at the University of Toronto

The University of Toronto is committed to equity, human rights, and respect for diversity. All members of the learning environment in this course should strive to create an atmosphere of mutual respect where all members of our community can express themselves, engage with each other, and respect one another's differences. U of T does not condone discrimination or harassment against any persons or communities.

Verification of Illness

Students who are absent from academic participation for **any reason** (e.g., COVID, cold, flu and other illness or injury, family situation) and who require consideration for missed academic work should report their absence through the online absence declaration. You must complete information at 2 locations:

First, you must complete the declaration available on **ACORN** **under the Profile and Settings menu.** Students should also advise their instructor of their absence.
You can formally declare an absence for one or multiple consecutive days using the Absence Declaration tool in ACORN. This tool is used to record any absences for courses you're taking during the current academic session.

- ACORN: *https://www.acorn.utoronto.ca/*
- To declare an absence
  https://help.acorn.utoronto.ca/blog/ufaqs/how-do-i-declare-an-absence/
- To edit an existing absence declaration

Secondly, you must also register your absence on the Department of Physical and Environmental Sciences "Self-declaration Absence Form" provided here: https://www.utsc.utoronto.ca/physsci/self-declaration-absence-form-0

Academic Integrity

All suspected cases of academic dishonesty will be investigated following procedures outlined in the Code of Behaviour on Academic Matters. If you have questions or concerns about what constitutes appropriate academic behaviour or appropriate research and citation methods, please reach out to me. Note that you are expected to seek out additional information on academic integrity from me or from other institutional resources (for example, the *University of Toronto website on Academic Integrity*).

Use of Generative Artificial Intelligence and similar technologies in this course

The use of generative artificial intelligence tools or apps for assignments in this course, including tools like ChatGPT, Gemini, Claude, Microsoft Copilot and other AI writing or coding assistants, is prohibited. The knowing use of generative artificial intelligence tools, including ChatGPT, Gemini, Claude, Microsoft Copilot and other AI writing and coding assistants, for the completion of, or to support the completion of, an examination, term test, assignment, or any other form of academic assessment, may be considered an academic offense in this course.

QUERCUS AND COURSE MATERIALS

Logging in to your Quercus Course Website

This course uses Quercus for its course website, which can be accessed via https://q.utoronto.ca.

Activating your UTORid and Password

If you need information on how to activate your UTORid and set your password for the first time, please go to http://www.utorid.utoronto.ca. Under the "First Time Users" area, click on "activate your UTORid" (if you are new to the university) or "create your UTORid" (if you are a returning student), then follow the instructions. If you have any issues, please contact the Student Help Desk at 416-978-HELP.

Communication *from* Course Instructor

If we need to contact you about an important change in the course, we will do so via the Quercus system. We suggest you consider downloading the "Canvas Student" app to your device(s) or, at a minimum, ensure your utoronto email address is properly linked for forwarding to whatever email service you actually check.

You are responsible for:

1. Ensuring you have a valid UofT email address that is properly entered in the ROSI/ACORN system.
2. Checking your UofT email account and/or Quercus communications on a regular basis as this is the primary means of professor-to-student communication outside of regular classroom hours.

Online Courses and Recording

Course videos and materials belong to your instructor, the University, and/or other sources depending on the specific facts of each situation and are protected by copyright. In this course, you are permitted to download session videos and materials for your own academic use, but you should not copy, input to a genAI system, or share or use them for any other purpose without the explicit permission of the instructor.

ADDITIONAL ASSITANCE

Writing and English Language

As well as the faculty writing support, please see English Language and writing support at University of Toronto: http://www.sgs.utoronto.ca/currentstudents/Pages/English-Language-and-Writing-Support.aspx Students have commented that they found this site extremely helpful for writing term papers.

The following are also useful:
Sylvan Barnett, *A Short Guide to Writing About Art*. 5-7th edition (New York: Harper-Collins, 1997)
William Strunk Jr., E.B. White. The Elements of Style (New York: MacMillan Publishing). Other information regarding writing support is provided on Blackboard.

Emergency Planning

Students are advised to consult the university's preparedness site for information and regular updates regarding procedures relating to emergency planning.
(http://www.preparedness.utoronto.ca)

Course Evaluation

I welcome your comments and concerns regarding the course or content directly by email or in person throughout the course. In addition, there is a formal procedure for submission of your comments regarding the course to the Department of Physical and Environmental Sciences at the completion of the course.

COURSE SCHEDULE S2026 – EESA11 ENVIRONMENTAL POLLUTION

Module	Week	Topic	Course Readings (R) & Videos (V) Additions to be made
1	May 4	Introduction to the Course • How the course works • Introduction too Environmental Pollution	R: *Earth Without Humans* (Holmes) V: Exxon Valdez Oil Spill: in the Wake of Disaster
2	May 11	The Science of Environmental Pollution • How pollutants move • Nature of pollutants	R: *Factors that led to the Walkerton Tragedy* (Salvadori et. Al) V: Walkerton Clean Water Centre: "25th Anniversary of the Walkerton Tragedy"
3	May 18	Air Pollution • Our atmosphere • Sources • Transport	R1: *Lessons learned from the Acid Rain Battles* R2: *The bittersweet story of how we stopped acid rain*
4	May 25	Pollution and Climate Change • Causes • Problems • Solutions?	R: *"United Nations Secretary-General's Call to Action on Extreme Heat "*– United Nations, July 25, 2024
5	June 1	Water Pollution • Sources and types • Solutions?	R1: *"7 Years Later …."* R2: *"Mount Polley disaster's toxic impact"* *V1: "Mount Polley Tailings Breach" V2: "Marking 10 years since Mount Polley Mine Disaster"*
N/A	June 8	**MIDTERM TEST (June 11)**	No Module Content this Week
No Class	June 17 to 20	***READING WEEK***	
6	June 22	Soil, Rock and Ground Water Pollutants	R: *"Toxic Peril - Love Canal"* V: *"Toxic Waste in the Neighbourhood"*
7	June 29	Agriculture and Pesticides • Practices and problems • Solutions	R: *"..how pesticides leach into the environment"* *V: "Do we really need pesticides?"*
8	July 6	Our waste – our planet • What have we done • What can we do?	R1: *"Waste Management in Remote Arctic Communities"* R2: *"Story Map – Arctic Conundrums"*
9	July 13	The Problem with Plastics • Hint: "they are a problem" • What can we do?	R1: *Solutions for plastic pollution*, Nature Geoscience (2023) V1: *"Plastic in the Great Lakes" V2: "Arctic Plastic"*
10	July 20	"Forever" Chemicals • What are they? • Where are they? • What do we know?	R: *"It's raining 'forever chemicals' across the great lakes. Why PFAS is now a public priority for elimination across Canada.* Canadian Environmental Law Association, November 4, 2021 V: "*What is Canada doing about forever chemicals?" About That CBC News*
11	July 27	Nuclear Waste and Radioactivity • Our "radioactive" legacy • Our "radioactive" future	V: "*CBC: Where will Canada put its forever nuclear waste dump"?*